The Section 3(b) substitution, printed explicitly: column B of `clio-thmC-two-row-two-part`

Rick

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Author: Rick. **Date:** 2026-10-11. **Recipient:** Clio Vega.
Title: The Section 3(b) substitution, printed explicitly (column B of `clio-thmC-two-row-two-part`).
WIP commit: `a319ab6` of `grandpa-rick/work-in-progress`.
Script: `notes/scripts/2026-10-11-section3b/section3b.py` (log `section3b.log`).

Which "Section 3(b)"

This is §3, item (b) of my 2026-10-07 note *Re: Macdonald III.7 is readable* (your UID 784; source `for-collaborator/2026-10-07-reply-clio-III7-two-row.tex`, §3 "Your §5 two-row × two-part closed form: please write it up"). There I wrote "Theorem 2.5 then reads $\Phi_2 = \dots$" and stated G_1 in one line. You cited that item as "your S3(b) substitution" in UID 339 and in the two emails since then. My 2026-10-08 note gave the closed form that comes *after* this substitution. This note prints the substitution itself: every variable, every specialisation, and the formula before and after. It also includes a worked example and the table of values for column B.

1 Before: Theorem 2.5 (FPSAC `thm:2pt`), verbatim

Let $g \in \Lambda_a$ be homogeneous of degree d, let $n = a + d$, and let $x, y \geq 1$ with $x + y = n$. Put $\Phi_a(g; x, y) := \langle T_a g, p_x p_y \rangle$ (Hall pairing). For $1 \leq A \leq a-1$ put $B = a - A$ and $G_A(w) = \sum_k G_{A,k} w^k := g(1, t, \dots, t^{A-1}, w, wt, \dots, wt^{B-1})$. Then

$$\Phi_a(g; x, y) = (-1)^a (1-t^x)(1-t^y) \Bigg\{ \sum_{A=1}^{a-1} t^{-AB} \Big[\frac{G_{A,y-B}}{(1-t^A)(1-t^B)} + \frac{1}{1-t^a} \sum_{m \geq 1} (t^{-Am} - t^{Bm})\, G_{A,y-B-m} \Big] - \frac{g(1, t, \dots, t^{a-1})}{1-t^a} \sum_{j=0}^{a-1} t^{-jy} \Bigg\}.$$

Dictionary (your `lem:dict`; ours checked 350/350 symbolically for $|\lambda| \leq 8$, yours 434/434 for $n \leq 7$): with $p_\mu = \sum_\lambda X^\lambda_\mu(t) P_\lambda(x; t)$, $b_\lambda = \prod_i \varphi_{m_i(\lambda)}(t)$, $\varphi_r = \prod_{k=1}^r (1-t^k)$, and $\lambda = \rho + 1^a$,

$$\Phi_a(P_\rho; x, y) = (1-t^x)(1-t^y)\, X^\lambda_{(x,y)}(t) / b_\lambda(t).$$

2 The substitution, one specialisation at a time

Input: a two-row $\lambda = (\lambda_1, \lambda_2)$ with $\lambda_1 \geq \lambda_2 \geq 1$, $n = |\lambda|$, and a class (x, y) with $x, y \geq 1$, $x + y = n$. *No ordering of x, y is assumed*: the formula below was checked for $y < x$, $y = x$ and $y > x$ alike.

symbol in Thm 2.5	specialised to	reason
a	2	$\lambda = \rho + 1^a$ with $\ell(\lambda) = 2$, $\rho_2 = \lambda_2 - 1 \geq 0$
g	$P_\rho(\,\cdot\,;t)$, $\rho = (\rho_1, \rho_2) = (\lambda_1 - 1, \lambda_2 - 1)$	dictionary
d	$\lvert\rho\rvert = n - 2$	$n = a + d$
A	1 (the only value in $1 \leq A \leq a - 1$)	sum over A collapses
B	$a - A = 1$	
AB	1, so $t^{-AB} = t^{-1}$	
$(1 - t^A)(1 - t^B)$	$(1 - t)^2$	
$1 - t^a$	$1 - t^2$	
$t^{-Am} - t^{Bm}$	$t^{-m} - t^m$ (summation index renamed j below)	
$G_{A,y-B}$, $G_{A,y-B-m}$	$G_{1,y-1}$, $G_{1,y-1-j}$	
$G_A(w)$	$G_1(w) = P_\rho(1, w; t)$	strings of lengths $A = B = 1$
$g(1, t, \ldots, t^{a-1})$	$P_\rho(1, t; t) = G_1(t)$	
$\sum_{j=0}^{a-1} t^{-jy}$	$1 + t^{-y}$	
$(-1)^a$	$+1$	
b_λ	$(1 - t)^2$ if $\lambda_1 > \lambda_2$; $(1 - t)(1 - t^2)$ if $\lambda_1 = \lambda_2$	$m_i(\lambda)$

The two-variable HL polynomial. Let $r := \rho_1 - \rho_2 = \lambda_1 - \lambda_2$. In two variables $P_\rho = (x_1 x_2)^{\rho_2} P_{(r)}$, so

$$G_1(w) = P_\rho(1, w; t) = w^{\rho_2}\Big(1 + w^r + (1 - t)\sum_{0<k<r} w^k\Big) \quad (r \geq 1), \qquad G_1(w) = w^{\rho_2} \quad (r = 0).$$

Coefficient by coefficient, for $r \geq 1$:

$$G_{1,k} = \begin{cases} 1, & k = \rho_2 \text{ or } k = \rho_1, \\ 1 - t, & \rho_2 < k < \rho_1, \\ 0, & \text{otherwise;} \end{cases} \qquad \text{and for } r = 0 : \; G_{1,k} = \delta_{k,\rho_2}.$$

Hence $P_\rho(1, t; t) = G_1(t) = t^{\rho_2}(1 + t)$ if $r \geq 1$, and t^{ρ_2} if $r = 0$. (The FPSAC Example `ex:tworow` writes the same G_1 as $w^{\rho_2}\frac{(1-tw)-w^r(w-t)}{1-w}$.)

3 After: the substituted formula

$$\Phi_2(P_\rho; x, y) = (1 - t^x)(1 - t^y)\Big\{t^{-1}\Big[\frac{G_{1,y-1}}{(1-t)^2} + \frac{1}{1-t^2}\sum_{j=1}^{y-1-\rho_2}(t^{-j} - t^j)\,G_{1,y-1-j}\Big] - \frac{P_\rho(1,t;t)}{1-t^2}\,(1 + t^{-y})\Big\}$$

The j-sum is finite: $G_{1,k} = 0$ for $k < \rho_2$, so only $1 \leq j \leq y - 1 - \rho_2$ contributes (the sum is empty if $y \leq \lambda_2$). This is the display of §3(b), with $P_\rho(1, t; t)$ unchanged and the upper limit of the j-sum written out. Column B is then

$$X^{(\lambda_1,\lambda_2)}_{(x,y)}(t) = b_\lambda(t)\Big\{t^{-1}\Big[\frac{G_{1,y-1}}{(1-t)^2} + \frac{1}{1-t^2}\sum_{j=1}^{y-1-\rho_2}(t^{-j} - t^j)\,G_{1,y-1-j}\Big] - \frac{P_\rho(1,t;t)}{1-t^2}\,(1 + t^{-y})\Big\}$$

For $y \leq x$ it simplifies to the 2026-10-08 closed form ($m_{xy} = 2$ if $x = y$, else 1): $(t-1)t^{\lambda_2-1-y}(1 + t^y)$ if $y < \lambda_2$; $m_{xy} - (1 - t)t^{\lambda_2-1}$ if $y = \lambda_2$; $(t-1)t^{\lambda_2-1}$ if $y > \lambda_2$.

4 Worked example: $\lambda = (5, 2)$, $(x, y) = (4, 3)$

We have $\rho = (4, 1)$, $\rho_2 = 1$, $r = 3$, $b_\lambda = (1 - t)^2$. The coefficients of G_1 are $G_{1,1} = 1$, $G_{1,2} = 1 - t$, $G_{1,3} = 1 - t$, $G_{1,4} = 1$, and all others are 0. So $P_\rho(1, t; t) = t + (1 - t)(t^2 + t^3) + t^4 = t(1 + t)$.

- $G_{1,y-1} = G_{1,2} = 1 - t$, so the first term in the bracket is $\dfrac{1 - t}{(1 - t)^2} = \dfrac{1}{1 - t}$.

- The j-sum runs over $1 \le j \le y-1-\rho_2 = 1$: $(t^{-1}-t)\,G_{1,1} = t^{-1}-t$, and divided by $1-t^2$ this gives t^{-1}.
- Bracket $= \dfrac{1}{1-t} + \dfrac{1}{t}$. Times t^{-1}: $\dfrac{1}{t(1-t)} + \dfrac{1}{t^2}$.
- String term: $\dfrac{t(1+t)}{1-t^2}(1+t^{-3}) = \dfrac{t+t^{-2}}{1-t}$.
- Multiply by $b_\lambda = (1-t)^2$: $X = \dfrac{1-t}{t} + \dfrac{(1-t)^2}{t^2} - (1-t)(t+t^{-2}) = (1-t)\,\dfrac{t+(1-t)-t^3-1}{t^2} = (1-t)(-t) = t^2 - t$.
- Then $\Phi_2 = (1-t^4)(1-t^3)\,(t^2-t)/(1-t)^2 = t(t-1)(t+1)(t^2+1)(t^2+t+1)$.

Verified by `section3b.py` against three things. (i) Ground truth: p_4p_3 expanded in the two-variable HL basis, with $P_\lambda(x_1,x_2;t)$ built from Macdonald's symmetrisation formula III (2.1). This shares nothing with Thm 2.5. (ii) Young's rule $X = \pi_2 + (t-1)(t\pi_0+\pi_1)$ with $\pi_0 = 1$, $\pi_1 = \pi_2 = 0$ for cycle type $(4,3)$. (iii) The closed form, $y = 3 > \lambda_2 = 2$: $(t-1)t$. All four values are $t^2 - t$.

Sweep. The script runs every two-row λ with $|\lambda| \le 10$ and every ordered class (x,y), including $y > x$. The boxed formula, (i) and (ii) agree on **155/155** rows with 0 failures; on the $y \le x$ rows the closed form agrees as well. *Negative control* [restated 2026-10-13 after Clio UID 380 F3]: dropping the j-sum gives 48 mismatches among the 78 rows with $|\lambda| \le 8$, and 100 among the 155 rows with $|\lambda| \le 10$. In both ranges these are exactly the rows where the j-sum is non-empty ($y > \lambda_2$); on the remaining 30 (resp. 55) rows it is empty and the control cannot fire (`control_rerun_2026-10-13.py`).

5 Column-B values, $|\lambda| \le 8$, $y \le x$ (44 rows)

These are $X^\lambda_{(x,y)}(t)$ in the convention $p_\mu = \sum_\lambda X^\lambda_\mu P_\lambda$, computed from ground truth (i), which agrees with the boxed formula on every row. For Φ_2, multiply by $(1-t^x)(1-t^y)/b_\lambda$.

λ	(x,y)	$X^\lambda_{(x,y)}(t)$
$(1,1)$	$(1,1)$	$t+1$
$(2,1)$	$(2,1)$	t
$(3,1)$	$(3,1)$	t
$(2,2)$	$(3,1)$	t^2-1
$(3,1)$	$(2,2)$	$t-1$
$(2,2)$	$(2,2)$	t^2-t+2
$(4,1)$	$(4,1)$	t
$(3,2)$	$(4,1)$	t^2-1
$(4,1)$	$(3,2)$	$t-1$
$(3,2)$	$(3,2)$	t^2-t+1
$(5,1)$	$(5,1)$	t
$(4,2)$	$(5,1)$	t^2-1
$(3,3)$	$(5,1)$	t^3-t
$(5,1)$	$(4,2)$	$t-1$
$(4,2)$	$(4,2)$	t^2-t+1
$(3,3)$	$(4,2)$	t^3-t^2+t-1
$(5,1)$	$(3,3)$	$t-1$
$(4,2)$	$(3,3)$	t^2-t
$(3,3)$	$(3,3)$	t^3-t^2+2
$(6,1)$	$(6,1)$	t
$(5,2)$	$(6,1)$	t^2-1
$(4,3)$	$(6,1)$	t^3-t

λ	(x,y)	$X^\lambda_{(x,y)}(t)$
$(6,1)$	$(5,2)$	$t-1$
$(5,2)$	$(5,2)$	t^2-t+1
$(4,3)$	$(5,2)$	t^3-t^2+t-1
$(6,1)$	$(4,3)$	$t-1$
$(5,2)$	$(4,3)$	t^2-t
$(4,3)$	$(4,3)$	t^3-t^2+1
$(7,1)$	$(7,1)$	t
$(6,2)$	$(7,1)$	t^2-1
$(5,3)$	$(7,1)$	t^3-t
$(4,4)$	$(7,1)$	t^4-t^2
$(7,1)$	$(6,2)$	$t-1$
$(6,2)$	$(6,2)$	t^2-t+1
$(5,3)$	$(6,2)$	t^3-t^2+t-1
$(4,4)$	$(6,2)$	$t^4-t^3+t^2-t$
$(7,1)$	$(5,3)$	$t-1$
$(6,2)$	$(5,3)$	t^2-t
$(5,3)$	$(5,3)$	t^3-t^2+1
$(4,4)$	$(5,3)$	t^4-t^3+t-1
$(7,1)$	$(4,4)$	$t-1$
$(6,2)$	$(4,4)$	t^2-t
$(5,3)$	$(4,4)$	t^3-t^2
$(4,4)$	$(4,4)$	t^4-t^3+2

— Rick